

Requirement Specification Document

Student Information Management System

Week 1 – Requirement Analysis and Student Database Module

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Project Title	Student Information Management System
Week	Week 1

1. Introduction:

1.1 Purpose

The purpose of this document is to analyze and specify the business requirements of the Student Information Management System. This system is designed to manage student records, course details, attendance, and academic performance efficiently.

1.2 Project Scope

The system will allow educational institutions to store and manage student information, course details, attendance records, and examination results. It will provide a centralized database for maintaining accurate student data and generating academic reports.

1.3 Intended Users

- Students – view personal details, attendance, and results.
- Faculty Members – manage attendance and student performance.
- Administrators – manage student records and generate reports.

2. Business Requirements

The following are the key business needs the system must satisfy:

1. Maintain accurate student records.
2. Store and manage course information.
3. Track student attendance.
4. Record examination marks and grades.
5. Generate academic and attendance reports.
6. Provide secure access to authorized users.

3. Functional Requirements

Functional requirements describe what the system must do.

ID	Requirement Description
FR-1	The system shall allow administrators to add new student records.
FR-2	The system shall allow administrators to update and delete student information.
FR-3	The system shall store course details for each student.
FR-4	The system shall allow faculty members to record attendance.
FR-5	The system shall store examination marks and grades.
FR-6	The system shall generate student performance reports.
FR-7	The system shall allow students to view their academic details.
FR-8	The system shall generate attendance reports for a selected period.

4. Non-Functional Requirements

- **Performance:** Student records should be retrieved within 2 seconds.
- **Security:** Student information must be protected from unauthorized access.
- **Reliability:** The database should maintain accurate and consistent data.
- **Usability:** The system interface should be simple and user-friendly.
- **Scalability:** The database should support increasing numbers of students and courses.

5. Student Database Module – Data Requirements

Attribute	Data Type	Description
student_id	INT (Primary Key)	Unique identifier for each student
full_name	VARCHAR(100)	Student's full name
gender	VARCHAR(100)	Student gender
date_of_birth	DATE	Student date of birth
email	VARCHAR(100)	Student email address
phone_number	VARCHAR(15)	Student contact number
address	VARCHAR(255)	Student residential address
course	VARCHAR(100)	Enrolled course
admission_date	DATE	Date of admission

6. Assumptions and Constraints

1. The system will be developed using a relational database such as MySQL, Oracle, or PostgreSQL.
2. Each student will have a unique Student ID.
3. Only authorized users can modify student records.
4. Student attendance and marks must be linked to valid student records.
5. This document covers Week 1 requirements only; additional modules will be developed in future weeks.

7. Conclusion

This document defines the initial business and functional requirements for the Student Information Management System, with a primary focus on the Student Database Module. It will serve as the foundation for database design, ER diagrams, schema creation, and future implementation phases.